

Problem Statement

Solo Innovator: **Maleto Hlolane**

Project Name: ClimaAware - Smart Indoor Weather Awareness System

Context & Problem Statement Background:

In many South African households, daily routines create a dangerous disconnect between indoor and outdoor conditions. After a hot bath, shower, or being inside a warm house, the human body feels warm and takes approximately 20-30 minutes to adjust to the real outside temperature. This thermal adaptation delay means people cannot accurately feel what the weather is truly like. The Problem: This is a critical health risk for vulnerable groups, especially children. On cold winter mornings when outside temperatures can be below 15°C, children who have just bathed run outside without jerseys because they still feel warm inside. On hot summer days above 30°C, they run outside and play too long in harsh sun because they do not realize the heat intensity. This indoor vs outdoor disconnect leads to preventable illnesses such as common colds, flu, respiratory infections, and heat-related discomfort. Parents currently have no simple, immediate warning at the exit door of the home. They rely on weather apps on phones, which are not visible to children and not checked every time someone leaves the house.

Validation of the Problem:

- To validate that this problem is real and not just my own experience, I conducted informal market research in my community in Tzaneen, Limpopo. I interviewed 2 mothers of young children:
- Mother 1: Said her child frequently catches a cold in winter after going outside without warm clothes after bathing.
- Mother 2 Said she checks her phone for weather, but her children do not, so they ignore her warnings.

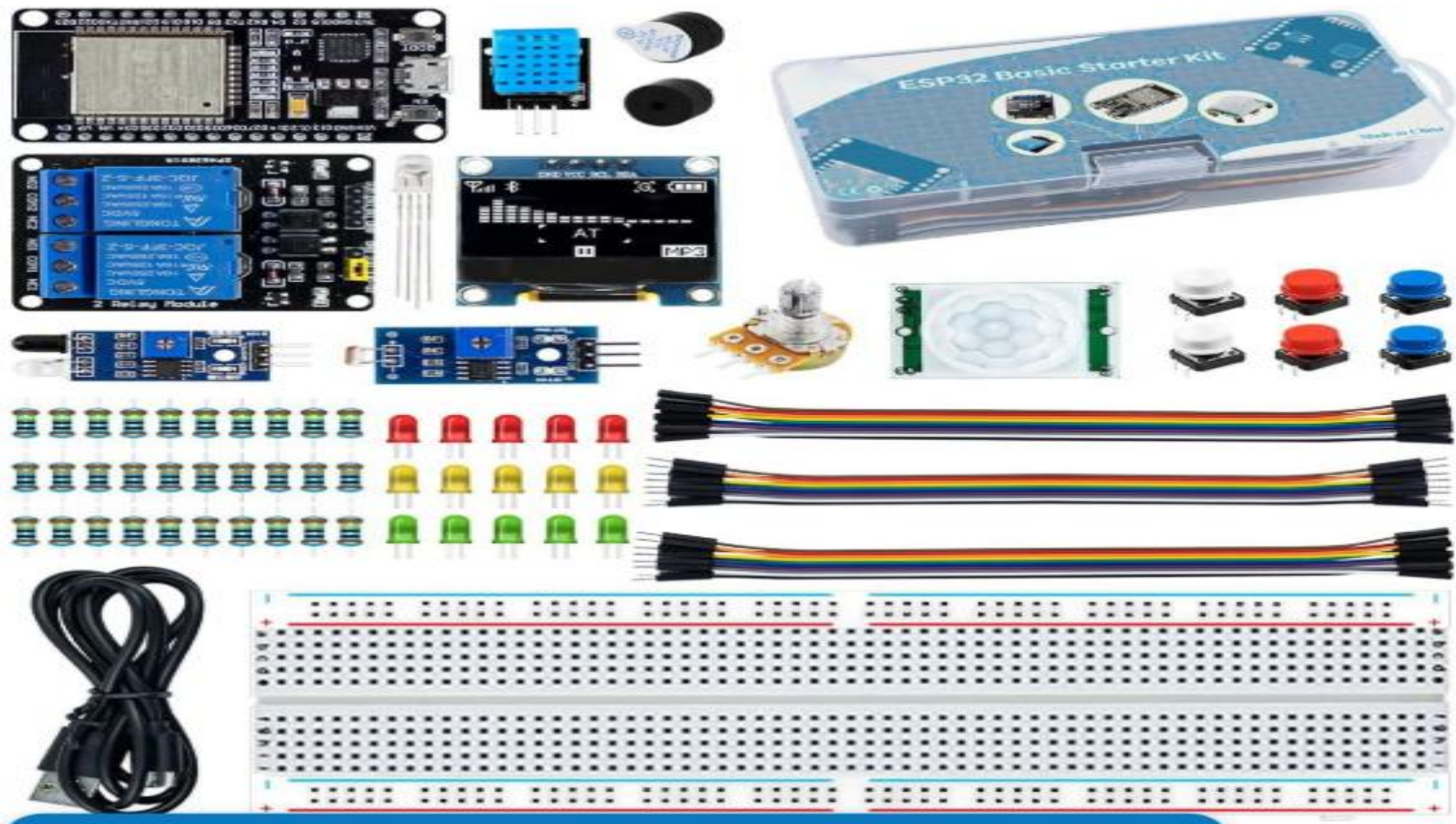
confirmed they would want a simple device at the door that tells the child what to do, like "Wear Jersey" or "Avoid Sun". This validates that the problem exists in real homes and there is a need for a low-cost, visual, child-friendly solution at the point of exit.

Solution and Marketability/Buildability

The solution is **ClimaAware**, a Smart Indoor Weather Awareness System that displays the real outdoor weather conditions inside the house. The system will use outdoor sensors and live weather data to detect the real-time outside temperature, whether it is cold, hot, humid, or windy. It will then present that information indoors through a smart display panel and ambient lighting system. If it is cold outside, the indoor ambient light will change to a cool blue tone and the display will clearly show the temperature with an icon of a jersey or jacket, indicating that warm clothes are needed. If it is extremely hot outside, the light will change to a warm orange or red tone and the display will show a sun icon with a warning to avoid direct sun. This visual system corrects the body's false feeling. Even if someone feels warm after taking a hot bath, they will see with their own eyes that it is actually cold outside. For children, the system will be installed at the main exit door and in the living area. It will use bright colors, large icons, and simple temperature numbers that are easy for children to understand, so before they go out to play, they instantly know what to wear and whether it is safe to play outside. In this way, the system makes the outdoor weather visible indoors, aligns human perception with reality, and prevents sickness caused by improper exposure to cold or heat.

Marketability/Buildability

- ClimaAware is marketable because it solves a real problem that causes children to get sick in Limpopo, where houses feel warm inside after a bath but it is actually cold outside, a problem validated by 2 mothers in Tzaneen. Its target market is large including parents, ECD centres, creches and primary schools in South Africa. Children are taught a simple rule without RGB, only two separate LEDs: if the GREEN LED is ON they can go outside and play, and if the RED LED is ON they must NOT go outside until parents allow them and give them a jersey or hat. It is cheap at R475.70 for the prototype using parts sourced from Takealot, with an ESP32, DHT22 outdoor sensor and 0.96 inch OLED, it needs no WiFi, no data and no subscription and runs on a normal 5V phone charger, and it is better than phone weather apps because it shows the real temperature of your yard at the exit door. It is buildable because it was built and tested in Tzaneen, Limpopo, South Africa in one day using off-the-shelf parts, with an outdoor DHT22 sensor wired to an indoor ESP32 that drives an OLED display showing temperature and jersey or sun icons and controls a GREEN LED for normal weather from 18 to 30 degrees and a RED LED for cold below 18 degrees or hot above 30 degrees plus a buzzer, fixed in a small wall box at the main exit door, with mass production cost below R500.



Basic Starter Kit for ESP32-32S WiFi I OT Development Board with Tutorial it

Total cost of implementation is R500

- <https://www.takealot.com/basic-starter-kit-for-esp32-32s-wifi-i-ot-development-board-with/PLID100195334>

Individual: Maleto Hlolane

